paul graham

on writing

essays, volume three

printed at home, october 2026

12 essays, 18,175 words. all of them are at paulgraham.com, free, as they should be.
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a printed book of essays
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01

writing, briefly

march 2005
paulgraham.com/writing44.html

(in the process of answering an email, i accidentally wrote a tiny essay about writing. i usually spend weeks on an essay. this one took 67 minutes—23 of writing, and 44 of rewriting.)

i think it's far more important to write well than most people realize. writing doesn't just communicate ideas; it generates them. if you're bad at writing and don't like to do it, you'll miss out on most of the ideas writing would have generated.

as for how to write well, here's the short version: write a bad version 1 as fast as you can; rewrite it over and over; cut ~~out~~ everything unnecessary; write in a conversational tone; develop a nose for bad writing, so you can see and fix it in yours; imitate writers you like; if you can't get started, tell someone what you plan to write about, then write down what you said; expect 80% of the ideas in an essay to happen after you start writing it, and 50% of those you start with to be wrong; be confident enough to cut; have friends you trust read your stuff and tell you which bits are confusing or drag; don't (always) make detailed outlines; mull ideas over for a few days before writing; carry a small notebook or scrap paper with you; start writing when you think of the first sentence; if a deadline forces you to start before that, just say the most important sentence first; write about stuff you like; don't try to sound impressive; don't hesitate to change the topic on the fly; use footnotes to contain digressions; use anaphora to knit sentences together; read your essays out loud to see (a) where you stumble over awkward phrases and (b) which bits are boring (the paragraphs you

dread reading); try to tell the reader something new and useful; work in fairly big quanta of time; when you restart, begin by rereading what you have so far; when you finish, leave yourself something easy to start with; accumulate notes for topics you plan to cover at the bottom of the file; don't feel obliged to cover any of them; write for a reader who won't read the essay as carefully as you do, just as pop songs are designed to sound ok on crappy car radios; if you say anything mistaken, fix it immediately; ask friends which sentence you'll regret most; go back and tone down harsh remarks; publish stuff online, because an audience makes you write more, and thus generate more ideas; print out drafts instead of just looking at them on the screen; use simple, germanic words; learn to distinguish surprises from digressions; learn to recognize the approach of an ending, and when one appears, grab it.

02

write simply

march 2021
paulgraham.com/simply.html

i try to write using ordinary words and simple sentences.
that kind of writing is easier to read, and the easier something is to read, the more deeply readers will engage with it. the less energy they expend on your prose, the more they'll have left for your ideas.
and the further they'll read. most readers' energy tends to flag part way through an article or essay. if the friction of reading is low enough, more keep going till the end.
there's an italian dish called *saltimbocca*, which means "leap into the mouth." my goal when writing might be called *saltintesta*: the ideas leap into your head and you barely notice the words that got them there.
it's too much to hope that writing could ever be pure ideas. you might not even want it to be. but for most writers, most of the time, that's the goal to aim for. the gap between most writing and pure ideas is not filled with poetry.
plus it's more considerate to write simply. when you write in a fancy way to impress people, you're making them do extra work just so you can seem cool. it's like trailing a long train behind you that readers have to carry.
and remember, if you're writing in english, that a lot of your readers won't be native english speakers. their understanding of ideas may be way ahead of their understanding of english. so you can't assume that writing about a difficult topic means you can use difficult words.
of course, fancy writing doesn't just conceal ideas. it can also conceal the lack of them. that's why some people write that way, to conceal the fact

that they have nothing to say. whereas writing simply keeps you honest. if you say nothing simply, it will be obvious to everyone, including you.
simple writing also lasts better. people reading your stuff in the future will be in much the same position as people from other countries reading it today. the culture and the language will have changed. it's not vain to care about that, any more than it's vain for a woodworker to build a chair to last.
indeed, lasting is not merely an accidental quality of chairs, or writing. it's a sign you did a good job.
but although these are all real advantages of writing simply, none of them are why i do it. the main reason i write simply is that it offends me not to. when i write a sentence that seems too complicated, or that uses unnecessarily intellectual words, it doesn't seem fancy to me. it seems clumsy.
there are of course times when you want to use a complicated sentence or fancy word for effect. but you should never do it by accident.
the other reason my writing ends up being simple is the way i do it. i write the first draft fast, then spend days editing it, trying to get everything just right. much of this editing is cutting, and that makes simple writing even simpler.

03

write like you talk

october 2015
paulgraham.com/talk.html

here's a simple trick for getting more people to read what you write: write in spoken language.
something comes over most people when they start writing. they write in a different language than they'd use if they were talking to a friend. the sentence structure and even the words are different. no one uses "pen" as a verb in spoken english. you'd feel like an idiot using "pen" instead of "write" in a conversation with a friend.
the last straw for me was a sentence i read a couple days ago:

> the mercurial spaniard himself declared: "after altamira, all is decadence."

it's from neil oliver's *a history of ancient britain*. i feel bad making an example of this book, because it's no worse than lots of others. but just imagine calling picasso "the mercurial spaniard" when talking to a friend. even one sentence of this would raise eyebrows in conversation. and yet people write whole books of it.
ok, so written and spoken language are different. does that make written language worse?
if you want people to read and understand what you write, yes. written language is more complex, which makes it more work to read. it's also more formal and distant, which gives the reader's attention permission to drift. but perhaps worst of all, the complex sentences and fancy words give

you, the writer, the false impression that you're saying more than you actually are.

you don't need complex sentences to express complex ideas. when specialists in some abstruse topic talk to one another about ideas in their field, they don't use sentences any more complex than they do when talking about what to have for lunch. they use different words, certainly. but even those they use no more than necessary. and in my experience, the harder the subject, the more informally experts speak. partly, i think, because they have less to prove, and partly because the harder the ideas you're talking about, the less you can afford to let language get in the way.

informal language is the athletic clothing of ideas.

i'm not saying spoken language always works best. poetry is as much music as text, so you can say things you wouldn't say in conversation. and there are a handful of writers who can get away with using fancy language in prose. and then of course there are cases where writers don't want to make it easy to understand what they're saying—in corporate announcements of bad news, for example, or at the more bogus end of the humanities. but for nearly everyone else, spoken language is better.

it seems to be hard for most people to write in spoken language. so perhaps the best solution is to write your first draft the way you usually would, then afterward look at each sentence and ask "is this the way i'd say this if i were talking to a friend?" if it isn't, imagine what you would say, and use that instead. after a while this filter will start to operate as you write. when you write something you wouldn't say, you'll hear the clank as it hits the page.

before i publish a new essay, i read it out loud and fix everything that doesn't sound like conversation. i even fix bits that are phonetically awkward; i don't know if that's necessary, but it doesn't cost much.

this trick may not always be enough. i've seen writing so far removed from spoken language that it couldn't be fixed sentence by sentence. for cases like that there's a more drastic solution. after writing the first draft, try exp-

laining to a friend what you just wrote. then replace the draft with what you said to your friend.

people often tell me how much my essays sound like me talking. the fact that this seems worthy of comment shows how rarely people manage to write in spoken language. otherwise everyone's writing would sound like them talking.

if you simply manage to write in spoken language, you'll be ahead of 95% of writers. and it's so easy to do: just don't let a sentence through unless it's the way you'd say it to a friend.

thanks to patrick collison and jessica livingston for reading drafts of this.

04

writes and write-nots

october 2024
paulgraham.com/writes.html

i'm usually reluctant to make predictions about technology, but i feel fairly confident about this one: in a couple decades there won't be many people who can write.

one of the strangest things you learn if you're a writer is how many people have trouble writing. doctors know how many people have a mole they're worried about; people who are good at setting up computers know how many people aren't; writers know how many people need help writing.

the reason so many people have trouble writing is that it's fundamentally difficult. to write well you have to think clearly, and thinking clearly is hard.

and yet writing pervades many jobs, and the more prestigious the job, the more writing it tends to require.

these two powerful opposing forces, the pervasive expectation of writing and the irreducible difficulty of doing it, create enormous pressure. this is why eminent professors often turn out to have resorted to plagiarism. the most striking thing to me about these cases is the pettiness of the thefts. the stuff they steal is usually the most mundane boilerplate — the sort of thing that anyone who was even halfway decent at writing could turn out with no effort at all. which means they're not even halfway decent at writing.

till recently there was no convenient escape valve for the pressure created by these opposing forces. you could pay someone to write for you, like jfk, or plagiarize, like mlk, but if you couldn't buy or steal words, you had to

write them yourself. and as a result nearly everyone who was expected to write had to learn how.

not anymore. ai has blown this world open. almost all pressure to write has dissipated. you can have ai do it for you, both in school and at work.

the result will be a world divided into writes and write-nots. there will still be some people who can write. some of us like it. but the middle ground between those who are good at writing and those who can't write at all will disappear. instead of good writers, ok writers, and people who can't write, there will just be good writers and people who can't write.

is that so bad? isn't it common for skills to disappear when technology makes them obsolete? there aren't many blacksmiths left, and it doesn't seem to be a problem.

yes, it's bad. the reason is something i mentioned earlier: writing is thinking. in fact there's a kind of thinking that can only be done by writing. you can't make this point better than leslie lamport did:

> if you're thinking without writing, you only think you're thinking.

so a world divided into writes and write-nots is more dangerous than it sounds. it will be a world of thinks and think-nots. i know which half i want to be in, and i bet you do too.

this situation is not unprecedented. in preindustrial times most people's jobs made them strong. now if you want to be strong, you work out. so there are still strong people, but only those who choose to be.

it will be the same with writing. there will still be smart people, but only those who choose to be.

thanks to jessica livingston, ben miller, and robert morris for reading drafts of this.

05

putting ideas into words

february 2022
paulgraham.com/words.html

writing about something, even something you know well, usually shows you that you didn't know it as well as you thought. putting ideas into words is a severe test. the first words you choose are usually wrong; you have to rewrite sentences over and over to get them exactly right. and your ideas won't just be imprecise, but incomplete too. half the ideas that end up in an essay will be ones you thought of while you were writing it. indeed, that's why i write them.

once you publish something, the convention is that whatever you wrote was what you thought before you wrote it. these were your ideas, and now you've expressed them. but you know this isn't true. you know that putting your ideas into words changed them. and not just the ideas you published. presumably there were others that turned out to be too broken to fix, and those you discarded instead.

it's not just having to commit your ideas to specific words that makes writing so exacting. the real test is reading what you've written. you have to pretend to be a neutral reader who knows nothing of what's in your head, only what you wrote. when he reads what you wrote, does it seem correct? does it seem complete? if you make an effort, you can read your writing as if you were a complete stranger, and when you do the news is usually bad. it takes me many cycles before i can get an essay past the stranger. but the stranger is rational, so you always can, if you ask him what he needs. if he's not satisfied because you failed to mention x or didn't qualify some sentence sufficiently, then you mention x or add more qualifications. happy

now? it may cost you some nice sentences, but you have to resign yourself to that. you just have to make them as good as you can and still satisfy the stranger.

this much, i assume, won't be that controversial. i think it will accord with the experience of anyone who has tried to write about anything nontrivial. there may exist people whose thoughts are so perfectly formed that they just flow straight into words. but i've never known anyone who could do this, and if i met someone who said they could, it would seem evidence of their limitations rather than their ability. indeed, this is a trope in movies: the guy who claims to have a plan for doing some difficult thing, and who when questioned further, taps his head and says "it's all up here." everyone watching the movie knows what that means. at best the plan is vague and incomplete. very likely there's some undiscovered flaw that invalidates it completely. at best it's a plan for a plan.

in precisely defined domains it's possible to form complete ideas in your head. people can play chess in their heads, for example. and mathematicians can do some amount of math in their heads, though they don't seem to feel sure of a proof over a certain length till they write it down. but this only seems possible with ideas you can express in a formal language. [1] arguably what such people are doing is putting ideas into words in their heads. i can to some extent write essays in my head. i'll sometimes think of a paragraph while walking or lying in bed that survives nearly unchanged in the final version. but really i'm writing when i do this. i'm doing the mental part of writing; my fingers just aren't moving as i do it. [2]

you can know a great deal about something without writing about it. can you ever know so much that you wouldn't learn more from trying to explain what you know? i don't think so. i've written about at least two subjects i know well — lisp hacking and startups — and in both cases i learned a lot from writing about them. in both cases there were things i didn't consciously realize till i had to explain them. and i don't think my experience was anomalous. a great deal of knowledge is unconscious, and experts

have if anything a higher proportion of unconscious knowledge than beginners.

i'm not saying that writing is the best way to explore all ideas. if you have ideas about architecture, presumably the best way to explore them is to build actual buildings. what i'm saying is that however much you learn from exploring ideas in other ways, you'll still learn new things from writing about them.

putting ideas into words doesn't have to mean writing, of course. you can also do it the old way, by talking. but in my experience, writing is the stricter test. you have to commit to a single, optimal sequence of words. less can go unsaid when you don't have tone of voice to carry meaning. and you can focus in a way that would seem excessive in conversation. i'll often spend 2 weeks on an essay and reread drafts 50 times. if you did that in conversation it would seem evidence of some kind of mental disorder. if you're lazy, of course, writing and talking are equally useless. but if you want to push yourself to get things right, writing is the steeper hill. [3]

the reason i've spent so long establishing this rather obvious point is that it leads to another that many people will find shocking. if writing down your ideas always makes them more precise and more complete, then no one who hasn't written about a topic has fully formed ideas about it. and someone who never writes has no fully formed ideas about anything nontrivial.

it feels to them as if they do, especially if they're not in the habit of critically examining their own thinking. ideas can feel complete. it's only when you try to put them into words that you discover they're not. so if you never subject your ideas to that test, you'll not only never have fully formed ideas, but also never realize it.

putting ideas into words is certainly no guarantee that they'll be right. far from it. but though it's not a sufficient condition, it is a necessary one.

notes

1 machinery and circuits are formal languages.

2 i thought of this sentence as i was walking down the street in palo alto.

3 there are two senses of talking to someone: a strict sense in which the conversation is verbal, and a more general sense in which it can take any form, including writing. in the limit case (e.g. seneca's letters), conversation in the latter sense becomes essay writing.

it can be very useful to talk (in either sense) with other people as you're writing something. but a verbal conversation will never be more exacting than when you're talking about something you're writing.

thanks to trevor blackwell, patrick collison, and robert morris for reading drafts of this.

06

how to write usefully

february 2020
paulgraham.com/useful.html

what should an essay be? many people would say persuasive. that's what a lot of us were taught essays should be. but i think we can aim for something more ambitious: that an essay should be useful.

to start with, that means it should be correct. but it's not enough merely to be correct. it's easy to make a statement correct by making it vague. that's a common flaw in academic writing, for example. if you know nothing at all about an issue, you can't go wrong by saying that the issue is a complex one, that there are many factors to be considered, that it's a mistake to take too simplistic a view of it, and so on.

though no doubt correct, such statements tell the reader nothing. useful writing makes claims that are as strong as they can be made without becoming false.

for example, it's more useful to say that pike's peak is near the middle of colorado than merely somewhere in colorado. but if i say it's in the exact middle of colorado, i've now gone too far, because it's a bit east of the middle.

precision and correctness are like opposing forces. it's easy to satisfy one if you ignore the other. the converse of vaporous academic writing is the bold, but false, rhetoric of demagogues. useful writing is bold, but true.

it's also two other things: it tells people something important, and that at least some of them didn't already know.

telling people something they didn't know doesn't always mean surprising them. sometimes it means telling them something they knew unconsciously

but had never put into words. in fact those may be the more valuable insights, because they tend to be more fundamental.

let's put them all together. useful writing tells people something true and important that they didn't already know, and tells them as unequivocally as possible.

notice these are all a matter of degree. for example, you can't expect an idea to be novel to everyone. any insight that you have will probably have already been had by at least one of the world's 7 billion people. but it's sufficient if an idea is novel to a lot of readers.

ditto for correctness, importance, and strength. in effect the four components are like numbers you can multiply together to get a score for usefulness. which i realize is almost awkwardly reductive, but nonetheless true.

———

how can you ensure that the things you say are true and novel and important? believe it or not, there is a trick for doing this. i learned it from my friend robert morris, who has a horror of saying anything dumb. his trick is not to say anything unless he's sure it's worth hearing. this makes it hard to get opinions out of him, but when you do, they're usually right.

translated into essay writing, what this means is that if you write a bad sentence, you don't publish it. you delete it and try again. often you abandon whole branches of four or five paragraphs. sometimes a whole essay.

you can't ensure that every idea you have is good, but you can ensure that every one you publish is, by simply not publishing the ones that aren't.

in the sciences, this is called publication bias, and is considered bad. when some hypothesis you're exploring gets inconclusive results, you're supposed to tell people about that too. but with essay writing, publication bias is the way to go.

my strategy is loose, then tight. i write the first draft of an essay fast, trying out all kinds of ideas. then i spend days rewriting it very carefully.

i've never tried to count how many times i proofread essays, but i'm sure there are sentences i've read 100 times before publishing them. when i proofread an essay, there are usually passages that stick out in an annoying

way, sometimes because they're clumsily written, and sometimes because i'm not sure they're true. the annoyance starts out unconscious, but after the tenth reading or so i'm saying "ugh, that part" each time i hit it. they become like briars that catch your sleeve as you walk past. usually i won't publish an essay till they're all gone — till i can read through the whole thing without the feeling of anything catching.

i'll sometimes let through a sentence that seems clumsy, if i can't think of a way to rephrase it, but i will never knowingly let through one that doesn't seem correct. you never have to. if a sentence doesn't seem right, all you have to do is ask why it doesn't, and you've usually got the replacement right there in your head.

this is where essayists have an advantage over journalists. you don't have a deadline. you can work for as long on an essay as you need to get it right. you don't have to publish the essay at all, if you can't get it right. mistakes seem to lose courage in the face of an enemy with unlimited resources. or that's what it feels like. what's really going on is that you have different expectations for yourself. you're like a parent saying to a child "we can sit here all night till you eat your vegetables." except you're the child too.

i'm not saying no mistake gets through. for example, i added condition (c) in "a way to detect bias" after readers pointed out that i'd omitted it. but in practice you can catch nearly all of them.

there's a trick for getting importance too. it's like the trick i suggest to young founders for getting startup ideas: to make something you yourself want. you can use yourself as a proxy for the reader. the reader is not completely unlike you, so if you write about topics that seem important to you, they'll probably seem important to a significant number of readers as well. importance has two factors. it's the number of people something matters to, times how much it matters to them. which means of course that it's not a rectangle, but a sort of ragged comb, like a riemann sum.

the way to get novelty is to write about topics you've thought about a lot. then you can use yourself as a proxy for the reader in this department too. anything you notice that surprises you, who've thought about the topic a

lot, will probably also surprise a significant number of readers. and here, as with correctness and importance, you can use the morris technique to ensure that you will. if you don't learn anything from writing an essay, don't publish it.

you need humility to measure novelty, because acknowledging the novelty of an idea means acknowledging your previous ignorance of it. confidence and humility are often seen as opposites, but in this case, as in many others, confidence helps you to be humble. if you know you're an expert on some topic, you can freely admit when you learn something you didn't know, because you can be confident that most other people wouldn't know it either.

the fourth component of useful writing, strength, comes from two things: thinking well, and the skillful use of qualification. these two counterbalance each other, like the accelerator and clutch in a car with a manual transmission. as you try to refine the expression of an idea, you adjust the qualification accordingly. something you're sure of, you can state baldly with no qualification at all, as i did the four components of useful writing. whereas points that seem dubious have to be held at arm's length with perhapses.

as you refine an idea, you're pushing in the direction of less qualification. but you can rarely get it down to zero. sometimes you don't even want to, if it's a side point and a fully refined version would be too long.

some say that qualifications weaken writing. for example, that you should never begin a sentence in an essay with "i think," because if you're saying it, then of course you think it. and it's true that "i think x" is a weaker statement than simply "x." which is exactly why you need "i think." you need it to express your degree of certainty.

but qualifications are not scalars. they're not just experimental error. there must be 50 things they can express: how broadly something applies, how you know it, how happy you are it's so, even how it could be falsified. i'm not going to try to explore the structure of qualification here. it's probably more complex than the whole topic of writing usefully. instead i'll just give

you a practical tip: don't underestimate qualification. it's an important skill in its own right, not just a sort of tax you have to pay in order to avoid saying things that are false. so learn and use its full range. it may not be fully half of having good ideas, but it's part of having them.

there's one other quality i aim for in essays: to say things as simply as possible. but i don't think this is a component of usefulness. it's more a matter of consideration for the reader. and it's a practical aid in getting things right; a mistake is more obvious when expressed in simple language. but i'll admit that the main reason i write simply is not for the reader's sake or because it helps get things right, but because it bothers me to use more or fancier words than i need to. it seems inelegant, like a program that's too long.

i realize florid writing works for some people. but unless you're sure you're one of them, the best advice is to write as simply as you can.

———

i believe the formula i've given you, importance + novelty + correctness + strength, is the recipe for a good essay. but i should warn you that it's also a recipe for making people mad.

the root of the problem is novelty. when you tell people something they didn't know, they don't always thank you for it. sometimes the reason people don't know something is because they don't want to know it. usually because it contradicts some cherished belief. and indeed, if you're looking for novel ideas, popular but mistaken beliefs are a good place to find them. every popular mistaken belief creates a dead zone of ideas around it that are relatively unexplored because they contradict it.

the strength component just makes things worse. if there's anything that annoys people more than having their cherished assumptions contradicted, it's having them flatly contradicted.

plus if you've used the morris technique, your writing will seem quite confident. perhaps offensively confident, to people who disagree with you. the reason you'll seem confident is that you are confident: you've cheated, by only publishing the things you're sure of. it will seem to people who try to

disagree with you that you never admit you're wrong. in fact you constantly admit you're wrong. you just do it before publishing instead of after.
and if your writing is as simple as possible, that just makes things worse. brevity is the diction of command. if you watch someone delivering unwelcome news from a position of inferiority, you'll notice they tend to use lots of words, to soften the blow. whereas to be short with someone is more or less to be rude to them.
it can sometimes work to deliberately phrase statements more weakly than you mean. to put "perhaps" in front of something you're actually quite sure of. but you'll notice that when writers do this, they usually do it with a wink.
i don't like to do this too much. it's cheesy to adopt an ironic tone for a whole essay. i think we just have to face the fact that elegance and curtness are two names for the same thing.
you might think that if you work sufficiently hard to ensure that an essay is correct, it will be invulnerable to attack. that's sort of true. it will be invulnerable to valid attacks. but in practice that's little consolation.
in fact, the strength component of useful writing will make you particularly vulnerable to misrepresentation. if you've stated an idea as strongly as you could without making it false, all anyone has to do is to exaggerate slightly what you said, and now it is false.
much of the time they're not even doing it deliberately. one of the most surprising things you'll discover, if you start writing essays, is that people who disagree with you rarely disagree with what you've actually written. instead they make up something you said and disagree with that.
for what it's worth, the countermove is to ask someone who does this to quote a specific sentence or passage you wrote that they believe is false, and explain why. i say "for what it's worth" because they never do. so although it might seem that this could get a broken discussion back on track, the truth is that it was never on track in the first place.
should you explicitly forestall likely misinterpretations? yes, if they're misinterpretations a reasonably smart and well-intentioned person might

make. in fact it's sometimes better to say something slightly misleading and then add the correction than to try to get an idea right in one shot. that can be more efficient, and can also model the way such an idea would be discovered.

but i don't think you should explicitly forestall intentional misinterpretations in the body of an essay. an essay is a place to meet honest readers. you don't want to spoil your house by putting bars on the windows to protect against dishonest ones. the place to protect against intentional misinterpretations is in end-notes. but don't think you can predict them all. people are as ingenious at misrepresenting you when you say something they don't want to hear as they are at coming up with rationalizations for things they want to do but know they shouldn't. i suspect it's the same skill.

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as with most other things, the way to get better at writing essays is to practice. but how do you start? now that we've examined the structure of useful writing, we can rephrase that question more precisely. which constraint do you relax initially? the answer is, the first component of importance: the number of people who care about what you write.

if you narrow the topic sufficiently, you can probably find something you're an expert on. write about that to start with. if you only have ten readers who care, that's fine. you're helping them, and you're writing. later you can expand the breadth of topics you write about.

the other constraint you can relax is a little surprising: publication. writing essays doesn't have to mean publishing them. that may seem strange now that the trend is to publish every random thought, but it worked for me. i wrote what amounted to essays in notebooks for about 15 years. i never published any of them and never expected to. i wrote them as a way of figuring things out. but when the web came along i'd had a lot of practice. incidentally, steve wozniak did the same thing. in high school he designed computers on paper for fun. he couldn't build them because he couldn't afford the components. but when intel launched 4k drams in 1975, he was ready.

how many essays are there left to write though? the answer to that question is probably the most exciting thing i've learned about essay writing. nearly all of them are left to write.

although the essay is an old form, it hasn't been assiduously cultivated. in the print era, publication was expensive, and there wasn't enough demand for essays to publish that many. you could publish essays if you were already well known for writing something else, like novels. or you could write book reviews that you took over to express your own ideas. but there was not really a direct path to becoming an essayist. which meant few essays got written, and those that did tended to be about a narrow range of subjects.

now, thanks to the internet, there's a path. anyone can publish essays online. you start in obscurity, perhaps, but at least you can start. you don't need anyone's permission.

it sometimes happens that an area of knowledge sits quietly for years, till some change makes it explode. cryptography did this to number theory. the internet is doing it to the essay.

the exciting thing is not that there's a lot left to write, but that there's a lot left to discover. there's a certain kind of idea that's best discovered by writing essays. if most essays are still unwritten, most such ideas are still undiscovered.

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notes

1 put railings on the balconies, but don't put bars on the windows.

2 even now i sometimes write essays that are not meant for publication. i wrote several to figure out what y combinator should do, and they were really helpful.

thanks to trevor blackwell, daniel gackle, jessica livingston, and robert morris for reading drafts of this.

07

good writing

may 2025
paulgraham.com/goodwriting.html

there are two senses in which writing can be good: it can sound good, and the ideas can be right. it can have nice, flowing sentences, and it can draw correct conclusions about important things. it might seem as if these two kinds of good would be unrelated, like the speed of a car and the color it's painted. and yet i don't think they are. i think writing that sounds good is more likely to be right.

so here we have the most exciting kind of idea: one that seems both preposterous and true. let's examine it. how can this possibly be true?

i know it's true from writing. you can't simultaneously optimize two unrelated things; when you push one far enough, you always end up sacrificing the other. and yet no matter how hard i push, i never find myself having to choose between the sentence that sounds best and the one that expresses an idea best. if i did, it would be frivolous to care how sentences sound. but in practice it feels the opposite of frivolous. fixing sentences that sound bad seems to help get the ideas right. [1]

by right i mean more than just true. getting the ideas right means developing them well — drawing the conclusions that matter most, and exploring each one to the right level of detail. so getting the ideas right is not just a matter of saying true things, but saying the right true things.

how could trying to make sentences sound good help you do that? the clue to the answer is something i noticed 30 years ago when i was doing the layout for my first book. sometimes when you're laying out text you have bad luck. for example, you get a section that runs one line longer than the

page. i don't know what ordinary typesetters do in this situation, but what i did was rewrite the section to make it a line shorter. you'd expect such an arbitrary constraint to make the writing worse. but i found, to my surprise, that it never did. i always ended up with something i liked better.

i don't think this was because my writing was especially careless. i think if you pointed to a random paragraph in anything written by anyone and told them to make it slightly shorter (or longer), they'd probably be able to come up with something better.

the best analogy for this phenomenon is when you shake a bin full of different objects. the shakes are arbitrary motions. or more precisely, they're not calculated to make any two specific objects fit more closely together. and yet repeated shaking inevitably makes the objects discover brilliantly clever ways of packing themselves. gravity won't let them become less tightly packed, so any change has to be a change for the better. [2]

so it is with writing. if you have to rewrite an awkward passage, you'll never do it in a way that makes it *less* true. you couldn't bear it, any more than gravity could bear things floating upward. so any change in the ideas has to be a change for the better.

it's obvious once you think about it. writing that sounds good is more likely to be right for the same reason that a well-shaken bin is more likely to be tightly packed. but there's something else going on as well. sounding good isn't just a random external force that leaves the ideas in an essay better off. it actually helps you to get them right.

the reason is that it makes the essay easier to read. it's less work to read writing that flows well. how does that help the writer? *because the writer is the first reader.* when i'm working on an essay, i spend far more time reading than writing. i'll reread some parts 50 or 100 times, replaying the thoughts in them and asking myself, like someone sanding a piece of wood, does anything catch? does anything feel wrong? and the easier the essay is to read, the easier it is to notice if something catches.

so yes, the two senses of good writing are connected in at least two ways. trying to make writing sound good makes you fix mistakes unconsciously,

and also helps you fix them consciously; it shakes the bin of ideas, and also makes mistakes easier to see. but now that we've dissolved one layer of preposterousness, i can't resist adding another. does sounding good do more than just help you get the ideas right? is writing that sounds good *inherently* more likely to be right? crazy as it may seem, i think that's true too.

obviously there's a connection at the level of individual words. there are lots of words in english that sound like what they mean, often in wonderfully subtle ways. glitter. round. scrape. prim. cavalcade. but the sound of good writing depends even more on the way you put words together, and there's a connection at that level too.

when writing sounds good, it's mostly because it has good rhythm. but the rhythm of good writing is not the rhythm of music, or the meter of verse. it's not so regular. if it were, it wouldn't be good, because the rhythm of good writing has to match the ideas in it, and ideas have all kinds of different shapes. sometimes they're simple and you just state them. but other times they're more subtle, and you need longer, more complicated sentences to tease out all the implications.

an essay is a cleaned up train of thought, in the same way dialogue is cleaned up conversation, and a train of thought has a natural rhythm. so when an essay sounds good, it's not merely because it has a pleasing rhythm, but because it has its natural one. which means you can use getting the rhythm right as a heuristic for getting the ideas right. and not just in principle: good writers do both simultaneously as a matter of course. often i don't even distinguish between the two problems. i just think ugh, this doesn't sound right; what do i mean to say here? [3]

the sound of writing turns out to be more like the shape of a plane than the color of a car. if it looks good, as kelly johnson used to say, it will fly well. this is only true of writing that's used to develop ideas, though. it doesn't apply when you have ideas in some other way and then write about them afterward — for example, if you build something, or conduct an experiment, and then write a paper about it. in such cases the ideas often live

more in the work than the writing, so the writing can be bad even though the ideas are good. the writing in textbooks and popular surveys can be bad for the same reason: the author isn't developing the ideas, merely describing other people's. it's only when you're writing to develop ideas that there's such a close connection between the two senses of doing it well.

ok, many people will be thinking, this seems plausible so far, but what about liars? is it not notoriously possible for a smooth-tongued liar to write something beautiful that's completely false?

it is, of course. but not without method acting. the way to write something beautiful and false is to begin by making yourself almost believe it. so just like someone writing something beautiful and true, you're presenting a perfectly-formed train of thought. the difference is the point where it attaches to the world. you're saying something that would be true if certain false premises were. if for some bizarre reason the number of jobs in a country were fixed, then immigrants really would be taking our jobs.

so it's not quite right to say that better sounding writing is more likely to be true. better sounding writing is more likely to be internally consistent. if the writer is honest, internal consistency and truth converge.

but while we can't safely conclude that beautiful writing is true, it's usually safe to conclude the converse: something that seems clumsily written will usually have gotten the ideas wrong too.

indeed, the two senses of good writing are more like two ends of the same thing. the connection between them is not a rigid one; the goodness of good writing is not a rod but a rope, with multiple overlapping connections running through it. but it's hard to move one end without moving the other. it's hard to be right without sounding right.

notes

1 the closest thing to an exception is when you have to go back and insert a new point into the middle of something you've written. this often messes up the flow, sometimes in ways you can never quite repair. but i think the ultimate source of this problem is that ideas are tree-shaped and essays are linear. you inevitably run into difficulties when you try to cram the former into the latter. frankly it's surprising how much you can get away with. but even so you sometimes have to resort to an endnote.

2 obviously if you shake the bin hard enough the objects in it can become less tightly packed. and similarly, if you imposed some huge external constraint on your writing, like using alternating one and two syllable words, the ideas would start to suffer.

3 bizarrely enough, this happened in the writing of this very paragraph. an earlier version shared several phrases in common with the preceding paragraph, and the repetition bugged me each time i reread it. when i got annoyed enough to fix it, i discovered that the repetition reflected a problem in the underlying ideas, and i fixed both simultaneously.

thanks to jessica livingston and courtenay pipkin for reading drafts of this.

08

the best essay

march 2024
paulgraham.com/best.html

despite its title this isn't meant to be the best essay. my goal here is to figure out what the best essay would be like.
it would be well-written, but you can write well about any topic. what made it special would be what it was about.
obviously some topics would be better than others. it probably wouldn't be about this year's lipstick colors. but it wouldn't be vaporous talk about elevated themes either. a good essay has to be surprising. it has to tell people something they don't already know.
the best essay would be on the most important topic you could tell people something surprising about.
that may sound obvious, but it has some unexpected consequences. one is that science enters the picture like an elephant stepping into a rowboat. for example, darwin first described the idea of natural selection in an essay written in 1844. talk about an important topic you could tell people something surprising about. if that's the test of a great essay, this was surely the best one written in 1844. and indeed, the best possible essay at any given time would usually be one describing the most important scientific or technological discovery it was possible to make. [1]
another unexpected consequence: i imagined when i started writing this that the best essay would be fairly timeless — that the best essay you could write in 1844 would be much the same as the best one you could write now. but in fact the opposite seems to be true. it might be true that the best painting would be timeless in this sense. but it wouldn't be impressive to

write an essay introducing natural selection now. the best essay *now* would be one describing a great discovery we didn't yet know about.

if the question of how to write the best possible essay reduces to the question of how to make great discoveries, then i started with the wrong question. perhaps what this exercise shows is that we shouldn't waste our time writing essays but instead focus on making discoveries in some specific domain. but i'm interested in essays and what can be done with them, so i want to see if there's some other question i could have asked.

there is, and on the face of it, it seems almost identical to the one i started with. instead of asking *what would the best essay be?* i should have asked *how do you write essays well?* though these seem only phrasing apart, their answers diverge. the answer to the first question, as we've seen, isn't really about essay writing. the second question forces it to be.

writing essays, at its best, is a way of discovering ideas. how do you do that well? how do you discover by writing?

an essay should ordinarily start with what i'm going to call a question, though i mean this in a very general sense: it doesn't have to be a question grammatically, just something that acts like one in the sense that it spurs some response.

how do you get this initial question? it probably won't work to choose some important-sounding topic at random and go at it. professional traders won't even trade unless they have what they call an *edge* — a convincing story about why in some class of trades they'll win more than they lose. similarly, you shouldn't attack a topic unless you have a way in — some new insight about it or way of approaching it.

you don't need to have a complete thesis; you just need some kind of gap you can explore. in fact, merely having questions about something other people take for granted can be edge enough.

if you come across a question that's sufficiently puzzling, it could be worth exploring even if it doesn't seem very momentous. many an important discovery has been made by pulling on a thread that seemed insignificant at first. how can they *all* be finches? [2]

once you've got a question, then what? you start thinking out loud about it. not literally out loud, but you commit to a specific string of words in response, as you would if you were talking. this initial response is usually mistaken or incomplete. writing converts your ideas from vague to bad. but that's a step forward, because once you can see the brokenness, you can fix it.

perhaps beginning writers are alarmed at the thought of starting with something mistaken or incomplete, but you shouldn't be, because this is why essay writing works. forcing yourself to commit to some specific string of words gives you a starting point, and if it's wrong, you'll see that when you reread it. at least half of essay writing is rereading what you've written and asking *is this correct and complete?* you have to be very strict when rereading, not just because you want to keep yourself honest, but because a gap between your response and the truth is often a sign of new ideas to be discovered.

the prize for being strict with what you've written is not just refinement. when you take a roughly correct answer and try to make it exactly right, sometimes you find that you can't, and that the reason is that you were depending on a false assumption. and when you discard it, the answer turns out to be completely different. [3]

ideally the response to a question is two things: the first step in a process that converges on the truth, and a source of additional questions (in my very general sense of the word). so the process continues recursively, as response spurs response. [4]

usually there are several possible responses to a question, which means you're traversing a tree. but essays are linear, not tree-shaped, which means you have to choose one branch to follow at each point. how do you choose? usually you should follow whichever offers the greatest combination of generality and novelty. i don't consciously rank branches this way; i just follow whichever seems most exciting; but generality and novelty are what make a branch exciting. [5]

if you're willing to do a lot of rewriting, you don't have to guess right. you can follow a branch and see how it turns out, and if it isn't good enough, cut it and backtrack. i do this all the time. in this essay i've already cut a 17-paragraph subtree, in addition to countless shorter ones. maybe i'll reattach it at the end, or boil it down to a footnote, or spin it off as its own essay; we'll see. [6]

in general you want to be quick to cut. one of the most dangerous temptations in writing (and in software and painting) is to keep something that isn't right, just because it contains a few good bits or cost you a lot of effort.

the most surprising new question being thrown off at this point is *does it really matter what the initial question is?* if the space of ideas is highly connected, it shouldn't, because you should be able to get from any question to the most valuable ones in a few hops. and we see evidence that it's highly connected in the way, for example, that people who are obsessed with some topic can turn any conversation toward it. but that only works if you know where you want to go, and you don't in an essay. that's the whole point. you don't want to be the obsessive conversationalist, or all your essays will be about the same thing. [7]

the other reason the initial question matters is that you usually feel somewhat obliged to stick to it. i don't think about this when i decide which branch to follow. i just follow novelty and generality. sticking to the question is enforced later, when i notice i've wandered too far and have to backtrack. but i think this is the optimal solution. you don't want the hunt for novelty and generality to be constrained in the moment. go with it and see what you get. [8]

since the initial question does constrain you, in the best case it sets an upper bound on the quality of essay you'll write. if you do as well as you possibly can on the chain of thoughts that follow from the initial question, the initial question itself is the only place where there's room for variation.

it would be a mistake to let this make you too conservative though, because you can't predict where a question will lead. not if you're doing

things right, because doing things right means making discoveries, and by definition you can't predict those. so the way to respond to this situation is not to be cautious about which initial question you choose, but to write a lot of essays. essays are for taking risks.

almost any question can get you a good essay. indeed, it took some effort to think of a sufficiently unpromising topic in the third paragraph, because any essayist's first impulse on hearing that the best essay couldn't be about x would be to try to write it. but if most questions yield good essays, only some yield great ones.

can we predict which questions will yield great essays? considering how long i've been writing essays, it's alarming how novel that question feels. one thing i like in an initial question is outrageousness. i love questions that seem naughty in some way — for example, by seeming counterintuitive or overambitious or heterodox. ideally all three. this essay is an example. writing about the best essay implies there is such a thing, which pseudo-intellectuals will dismiss as reductive, though it follows necessarily from the possibility of one essay being better than another. and thinking about how to do something so ambitious is close enough to doing it that it holds your attention.

i like to start an essay with a gleam in my eye. this could be just a taste of mine, but there's one aspect of it that probably isn't: to write a really good essay on some topic, you have to be interested in it. a good writer can write well about anything, but to stretch for the novel insights that are the raison d'etre of the essay, you have to care.

if caring about it is one of the criteria for a good initial question, then the optimal question varies from person to person. it also means you're more likely to write great essays if you care about a lot of different things. the more curious you are, the greater the probable overlap between the set of things you're curious about and the set of topics that yield great essays.

what other qualities would a great initial question have? it's probably good if it has implications in a lot of different areas. and i find it's a good sign if it's one that people think has already been thoroughly explored. but the

truth is that i've barely thought about how to choose initial questions, because i rarely do it. i rarely *choose* what to write about; i just start thinking about something, and sometimes it turns into an essay.

am i going to stop writing essays about whatever i happen to be thinking about and instead start working my way through some systematically generated list of topics? that doesn't sound like much fun. and yet i want to write good essays, and if the initial question matters, i should care about it. perhaps the answer is to go one step earlier: to write about whatever pops into your head, but try to ensure that what pops into your head is good. indeed, now that i think about it, this has to be the answer, because a mere list of topics wouldn't be any use if you didn't have edge with any of them. to start writing an essay, you need a topic plus some initial insight about it, and you can't generate those systematically. if only. [9]

you can probably cause yourself to have more of them, though. the quality of the ideas that come out of your head depends on what goes in, and you can improve that in two dimensions, breadth and depth.

you can't learn everything, so getting breadth implies learning about topics that are very different from one another. when i tell people about my book-buying trips to hay and they ask what i buy books about, i usually feel a bit sheepish answering, because the topics seem like a laundry list of unrelated subjects. but perhaps that's actually optimal in this business.

you can also get ideas by talking to people, by doing and building things, and by going places and seeing things. i don't think it's important to talk to new people so much as the sort of people who make you have new ideas. i get more new ideas after talking for an afternoon with robert morris than from talking to 20 new smart people. i know because that's what a block of office hours at y combinator consists of.

while breadth comes from reading and talking and seeing, depth comes from doing. the way to really learn about some domain is to have to solve problems in it. though this could take the form of writing, i suspect that to be a good essayist you also have to do, or have done, some other kind of

nt. you could spend half your time working on something else and be net ahead, so long as it was hard.

i'm not proposing that as a recipe so much as an encouragement to those already doing it. if you've spent all your life so far working on other things, you're already halfway there. though of course to be good at writing you have to like it, and if you like writing you'd probably have spent at least some time doing it.

everything i've said about initial questions applies also to the questions you encounter in writing the essay. they're the same thing; every subtree of an essay is usually a shorter essay, just as every subtree of a calder mobile is a smaller mobile. so any technique that gets you good initial questions also gets you good whole essays.

at some point the cycle of question and response reaches what feels like a natural end. which is a little suspicious; shouldn't every answer suggest more questions? i think what happens is that you start to feel sated. once you've covered enough interesting ground, you start to lose your appetite for new questions. which is just as well, because the reader is probably feeling sated too. and it's not lazy to stop asking questions, because you could instead be asking the initial question of a new essay.

that's the ultimate source of drag on the connectedness of ideas: the discoveries you make along the way. if you discover enough starting from question a, you'll never make it to question b. though if you keep writing essays you'll gradually fix this problem by burning off such discoveries. so bizarrely enough, writing lots of essays makes it as if the space of ideas were more highly connected.

when a subtree comes to an end, you can do one of two things. you can either stop, or pull the cubist trick of laying separate subtrees end to end by returning to a question you skipped earlier. usually it requires some sleight of hand to make the essay flow continuously at this point, but not this time. this time i actually need an example of the phenomenon. for example, we discovered earlier that the best possible essay wouldn't usually be

timeless in the way the best painting would. this seems surprising enough to be worth investigating further.

there are two senses in which an essay can be timeless: to be about a matter of permanent importance, and always to have the same effect on readers. with art these two senses blend together. art that looked beautiful to the ancient greeks still looks beautiful to us. but with essays the two senses diverge, because essays teach, and you can't teach people something they already know. natural selection is certainly a matter of permanent importance, but an essay explaining it couldn't have the same effect on us that it would have had on darwin's contemporaries, precisely because his ideas were so successful that everyone already knows about them. [10]

i imagined when i started writing this that the best possible essay would be timeless in the stricter, evergreen sense: that it would contain some deep, timeless wisdom that would appeal equally to aristotle and feynman. that doesn't seem to be true. but if the best possible essay wouldn't usually be timeless in this stricter sense, what would it take to write essays that were? the answer to that turns out to be very strange: to be the evergreen kind of timeless, an essay has to be ineffective, in the sense that its discoveries aren't assimilated into our shared culture. otherwise there will be nothing new in it for the second generation of readers. if you want to surprise readers not just now but in the future as well, you have to write essays that won't stick — essays that, no matter how good they are, won't become part of what people in the future learn before they read them. [11]

i can imagine several ways to do that. one would be to write about things people never learn. for example, it's a long-established pattern for ambitious people to chase after various types of prizes, and only later, perhaps too late, to realize that some of them weren't worth as much as they thought. if you write about that, you can be confident of a conveyor belt of future readers to be surprised by it.

ditto if you write about the tendency of the inexperienced to overdo things — of young engineers to produce overcomplicated solutions, for example.

there are some kinds of mistakes people never learn to avoid except by making them. any of those should be a timeless topic.

sometimes when we're slow to grasp things it's not just because we're obtuse or in denial but because we've been deliberately lied to. there are a lot of things adults lie to kids about, and when you reach adulthood, they don't take you aside and hand you a list of them. they don't remember which lies they told you, and most were implicit anyway. so contradicting such lies will be a source of surprises for as long as adults keep telling them.

sometimes it's systems that lie to you. for example, the educational systems in most countries train you to win by hacking the test. but that's not how you win at the most important real-world tests, and after decades of training, this is hard for new arrivals in the real world to grasp. helping them overcome such institutional lies will work as long as the institutions remain broken. [12]

another recipe for timelessness is to write about things readers already know, but in much more detail than can be transmitted culturally. "everyone knows," for example, that it can be rewarding to have kids. but till you have them you don't know precisely what forms that takes, and even then much of what you know you may never have put into words.

i've written about all these kinds of topics. but i didn't do it in a deliberate attempt to write essays that were timeless in the stricter sense. and indeed, the fact that this depends on one's ideas not sticking suggests that it's not worth making a deliberate attempt to. you should write about topics of timeless importance, yes, but if you do such a good job that your conclusions stick and future generations find your essay obvious instead of novel, so much the better. you've crossed into darwin territory.

writing about topics of timeless importance is an instance of something even more general, though: breadth of applicability. and there are more kinds of breadth than chronological — applying to lots of different fields, for example. so breadth is the ultimate aim.

i already aim for it. breadth and novelty are the two things i'm always chasing. but i'm glad i understand where timelessness fits.

i understand better where a lot of things fit now. this essay has been a kind of tour of essay writing. i started out hoping to get advice about topics; if you assume good writing, the only thing left to differentiate the best essay is its topic. and i did get advice about topics: discover natural selection. yeah, that would be nice. but when you step back and ask what's the best you can do short of making some great discovery like that, the answer turns out to be about procedure. ultimately the quality of an essay is a function of the ideas discovered in it, and the way you get them is by casting a wide net for questions and then being very exacting with the answers.

the most striking feature of this map of essay writing are the alternating stripes of inspiration and effort required. the questions depend on inspiration, but the answers can be got by sheer persistence. you don't have to get an answer right the first time, but there's no excuse for not getting it right eventually, because you can keep rewriting till you do. and this is not just a theoretical possibility. it's a pretty accurate description of the way i work. i'm rewriting as we speak.

but although i wish i could say that writing great essays depends mostly on effort, in the limit case it's inspiration that makes the difference. in the limit case, the questions are the harder thing to get. that pool has no bottom. how to get more questions? that is the most important question of all.

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notes

1 there might be some resistance to this conclusion on the grounds that some of these discoveries could only be understood by a small number of readers. but you get into all sorts of difficulties if you want to disqualify essays on this account. how do you decide where the cutoff should be? if a virus kills off everyo-

ne except a handful of people sequestered at los alamos, could an essay that had been disqualified now be eligible? etc.

darwin's 1844 essay was derived from an earlier version written in 1839. extracts from it were published in 1858.

2 when you find yourself very curious about an apparently minor question, that's an exciting sign. evolution has designed you to pay attention to things that matter. so when you're very curious about something random, that could mean you've unconsciously noticed it's less random than it seems.

3 corollary: if you're not intellectually honest, your writing won't just be biased, but also boring, because you'll miss all the ideas you'd have discovered if you pushed for the truth.

4 sometimes this process begins before you start writing. sometimes you've already figured out the first few things you want to say. schoolchildren are often taught they should decide *everything* they want to say, and write this down as an outline before they start writing the essay itself. maybe that's a good way to get them started — or not, i don't know — but it's antithetical to the spirit of essay writing. the more detailed your outline, the less your ideas can benefit from the sort of discovery that essays are for.

5 the problem with this type of "greedy" algorithm is that you can end up on a local maximum. if the most valuable question is preceded by a boring one, you'll overlook it. but i can't imagine a better strategy. there's no lookahead except by writing. so use a greedy algorithm and a lot of time.

6 i ended up reattaching the first 5 of the 17 paragraphs, and discarding the rest.

7 stephen fry confessed to making use of this phenomenon when taking exams at oxford. he had in his head a standard essay about some general literary topic, and he would find a way to turn the exam question toward it and then just reproduce it again.

strictly speaking it's the graph of ideas that would be highly connected, not the space, but that usage would confuse people who don't know graph theory, whereas people who do know it will get what i mean if i say "space".

8 too far doesn't depend just on the distance from the original topic. it's more like that distance divided by the value of whatever i've discovered in the subtree.

9 or can you? i should try writing about this. even if the chance of succeeding is small, the expected value is huge.

10 there was a vogue in the 20th century for saying that the purpose of art was also to teach. some artists tried to justify their work by explaining that their goal was not to produce something good, but to challenge our preconceptions about art. and to be fair, art can teach somewhat. the ancient greeks' naturalistic sculptures represented a new idea, and must have been extra exciting to contemporaries on that account. but they still look good to us.

11 bertrand russell caused huge controversy in the early 20th century with his ideas about "trial marriage." but they make boring reading now, because they prevailed. "trial marriage" is what we call "dating."

12 if you'd asked me 10 years ago, i'd have predicted that schools would continue to teach hacking the test for centuries. but now it seems plausible that students will soon be taught individually by ais, and that exams will be replaced by ongoing, invisible micro-assessments.

thanks to sam altman, trevor blackwell, jessica livingston, robert morris, courtenay pipkin, and harj taggar for reading drafts of this.

09

the shape of the essay field

june 2025
paulgraham.com/field.html

an essay has to tell people something they don't already know. but there are three different reasons people might not know something, and they yield three very different kinds of essays.

one reason people won't know something is if it's not important to know. that doesn't mean it will make a bad essay. for example, you might write a good essay about a particular model of car. readers would learn something from it. it would add to their picture of the world. for a handful of readers it might even spur some kind of epiphany. but unless this is a very unusual car it's not critical for everyone to know about it. [1]

if something isn't important to know, there's no answer to the question of why people don't know it. not knowing random facts is the default. but if you're going to write about things that *are* important to know, you have to ask why your readers don't already know them. is it because they're inexperienced, or because they're obtuse?

so the three reasons readers might not already know what you tell them are (a) that it's not important, (b) that they're obtuse, or (c) that they're inexperienced.

the reason i did this breakdown was to get at the following fact, which might have seemed controversial if i'd led with it, but should be obvious now. if you're writing for smart people about important things, you're writing for the young.

or more precisely, that's where you'll have the most effect. whatever you say should also be at least somewhat novel to you, however old you are.

it's not an essay otherwise, because an essay is something you write to figure something out. but whatever you figure out will presumably be more of a surprise to younger readers than it is to you.

there's a continuum of surprise. at one extreme, something you read can change your whole way of thinking. *the selfish gene* did this to me. it was like suddenly seeing the other interpretation of an ambiguous image: you can treat genes rather than organisms as the protagonists, and evolution becomes easier to understand when you do. at the other extreme, writing merely puts into words something readers were already thinking — or thought they were.

the impact of an essay is how much it changes readers' thinking multiplied by the importance of the topic. but it's hard to do well at both. it's hard to have big new ideas about important topics. so in practice there's a tradeoff: you can change readers' thinking a lot about moderately important things, or change it a little about very important ones. but with younger readers the tradeoff shifts. there's more room to change their thinking, so there's a bigger payoff for writing about important things.

the tradeoff isn't a conscious one, at least not for me. it's more like a kind of gravitational field that writers work in. but every essayist works in it, whether they realize it or not.

this seems obvious once you state it, but it took me a long time to understand. i knew i wanted to write for smart people about important topics. i noticed empirically that i seemed to be writing for the young. but it took me years to understand that the latter was an automatic consequence of the former. in fact i only really figured it out as i was writing this essay.

now that i know it, should i change anything? i don't think so. in fact seeing the shape of the field that writers work in has reminded me that i'm not optimizing for returns in it. i'm not trying to surprise readers of any particular age; i'm trying to surprise myself.

the way i usually decide what to write about is by following curiosity. i notice something new and dig into it. it would probably be a mistake to change that. but seeing the shape of the essay field has set me thinking.

what would surprise young readers? which important things do people tend to learn late? interesting question. i should think about that.

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notes

1 it's hard to write a really good essay about an unimportant topic, though, because a really good essayist will inevitably draw the topic into deeper waters. e. b. white could write an essay about how to boil potatoes that ended up being full of timeless wisdom. in which case, of course, it wouldn't really be about how to boil potatoes; that would just have been the starting point.

thanks to jessica livingston and michael nielsen for reading drafts of this.

10

the age of the essay

september 2004
paulgraham.com/essay.html

remember the essays you had to write in high school? topic sentence, introductory paragraph, supporting paragraphs, conclusion. the conclusion being, say, that ahab in *moby dick* was a christ-like figure.
oy. so i'm going to try to give the other side of the story: what an essay really is, and how you write one. or at least, how i write one.
mods
the most obvious difference between real essays and the things one has to write in school is that real essays are not exclusively about english literature. certainly schools should teach students how to write. but due to a series of historical accidents the teaching of writing has gotten mixed together with the study of literature. and so all over the country students are writing not about how a baseball team with a small budget might compete with the yankees, or the role of color in fashion, or what constitutes a good dessert, but about symbolism in dickens.
with the result that writing is made to seem boring and pointless. who cares about symbolism in dickens? dickens himself would be more interested in an essay about color or baseball.
how did things get this way? to answer that we have to go back almost a thousand years. around 1100, europe at last began to catch its breath after centuries of chaos, and once they had the luxury of curiosity they rediscovered what we call "the classics." the effect was rather as if we were visited by beings from another solar system. these earlier civilizations were so much more sophisticated that for the next several centuries the main work

of european scholars, in almost every field, was to assimilate what they knew.

during this period the study of ancient texts acquired great prestige. it seemed the essence of what scholars did. as european scholarship gained momentum it became less and less important; by 1350 someone who wanted to learn about science could find better teachers than aristotle in his own era.[1] but schools change slower than scholarship. in the 19th century the study of ancient texts was still the backbone of the curriculum.

the time was then ripe for the question: if the study of ancient texts is a valid field for scholarship, why not modern texts? the answer, of course, is that the original raison d'etre of classical scholarship was a kind of intellectual archaeology that does not need to be done in the case of contemporary authors. but for obvious reasons no one wanted to give that answer. the archaeological work being mostly done, it implied that those studying the classics were, if not wasting their time, at least working on problems of minor importance.

and so began the study of modern literature. there was a good deal of resistance at first. the first courses in english literature seem to have been offered by the newer colleges, particularly american ones. dartmouth, the university of vermont, amherst, and university college, london taught english literature in the 1820s.

but harvard didn't have a professor of english literature until 1876, and oxford not till 1885. (oxford had a chair of chinese before it had one of english.)[2]

what tipped the scales, at least in the us, seems to have been the idea that professors should do research as well as teach. this idea (along with the phd, the department, and indeed the whole concept of the modern university) was imported from germany in the late 19th century. beginning at johns hopkins in 1876, the new model spread rapidly.

writing was one of the casualties. colleges had long taught english composition. but how do you do research on composition? the professors who taught math could be required to do original math, the professors who

taught history could be required to write scholarly articles about history, but what about the professors who taught rhetoric or composition? what should they do research on? the closest thing seemed to be english literature.[3]

and so in the late 19th century the teaching of writing was inherited by english professors. this had two drawbacks: (a) an expert on literature need not himself be a good writer, any more than an art historian has to be a good painter, and (b) the subject of writing now tends to be literature, since that's what the professor is interested in.

high schools imitate universities. the seeds of our miserable high school experiences were sown in 1892, when the national education association "formally recommended that literature and composition be unified in the high school course."[4] the 'riting component of the 3 rs then morphed into english, with the bizarre consequence that high school students now had to write about english literature-- to write, without even realizing it, imitations of whatever english professors had been publishing in their journals a few decades before.

it's no wonder if this seems to the student a pointless exercise, because we're now three steps removed from real work: the students are imitating english professors, who are imitating classical scholars, who are merely the inheritors of a tradition growing out of what was, 700 years ago, fascinating and urgently needed work.

no defense

the other big difference between a real essay and the things they make you write in school is that a real essay doesn't take a position and then defend it. that principle, like the idea that we ought to be writing about literature, turns out to be another intellectual hangover of long forgotten origins.

it's often mistakenly believed that medieval universities were mostly seminaries. in fact they were more law schools. and at least in our tradition lawyers are advocates, trained to take either side of an argument and make as good a case for it as they can. whether cause or effect, this spirit pervaded early universities. the study of rhetoric, the art of arguing persuasively,

was a third of the undergraduate curriculum.[5] and after the lecture the most common form of discussion was the disputation. this is at least nominally preserved in our present-day thesis defense: most people treat the words thesis and dissertation as interchangeable, but originally, at least, a thesis was a position one took and the dissertation was the argument by which one defended it.

defending a position may be a necessary evil in a legal dispute, but it's not the best way to get at the truth, as i think lawyers would be the first to admit. it's not just that you miss subtleties this way. the real problem is that you can't change the question.

and yet this principle is built into the very structure of the things they teach you to write in high school. the topic sentence is your thesis, chosen in advance, the supporting paragraphs the blows you strike in the conflict, and the conclusion-- uh, what is the conclusion? i was never sure about that in high school. it seemed as if we were just supposed to restate what we said in the first paragraph, but in different enough words that no one could tell. why bother? but when you understand the origins of this sort of "essay," you can see where the conclusion comes from. it's the concluding remarks to the jury.

good writing should be convincing, certainly, but it should be convincing because you got the right answers, not because you did a good job of arguing. when i give a draft of an essay to friends, there are two things i want to know: which parts bore them, and which seem unconvincing. the boring bits can usually be fixed by cutting. but i don't try to fix the unconvincing bits by arguing more cleverly. i need to talk the matter over.

at the very least i must have explained something badly. in that case, in the course of the conversation i'll be forced to come up a with a clearer explanation, which i can just incorporate in the essay. more often than not i have to change what i was saying as well. but the aim is never to be convincing per se. as the reader gets smarter, convincing and true become identical, so if i can convince smart readers i must be near the truth.

the sort of writing that attempts to persuade may be a valid (or at least inevitable) form, but it's historically inaccurate to call it an essay. an essay is something else.

trying

to understand what a real essay is, we have to reach back into history again, though this time not so far. to michel de montaigne, who in 1580 published a book of what he called "essais." he was doing something quite different from what lawyers do, and the difference is embodied in the name. *essayer* is the french verb meaning "to try" and an *essai* is an attempt. an essay is something you write to try to figure something out. figure out what? you don't know yet. and so you can't begin with a thesis, because you don't have one, and may never have one. an essay doesn't begin with a statement, but with a question. in a real essay, you don't take a position and defend it. you notice a door that's ajar, and you open it and walk in to see what's inside.

if all you want to do is figure things out, why do you need to write anything, though? why not just sit and think? well, there precisely is montaigne's great discovery. expressing ideas helps to form them. indeed, helps is far too weak a word. most of what ends up in my essays i only thought of when i sat down to write them. that's why i write them.

in the things you write in school you are, in theory, merely explaining yourself to the reader. in a real essay you're writing for yourself. you're thinking out loud.

but not quite. just as inviting people over forces you to clean up your apartment, writing something that other people will read forces you to think well. so it does matter to have an audience. the things i've written just for myself are no good. they tend to peter out. when i run into difficulties, i find i conclude with a few vague questions and then drift off to get a cup of tea.

many published essays peter out in the same way. particularly the sort written by the staff writers of newsmagazines. outside writers tend to supply editorials of the defend-a-position variety, which make a beeline toward a

rousing (and foreordained) conclusion. but the staff writers feel obliged to write something "balanced." since they're writing for a popular magazine, they start with the most radioactively controversial questions, from which-- because they're writing for a popular magazine-- they then proceed to recoil in terror. abortion, for or against? this group says one thing. that group says another. one thing is certain: the question is a complex one. (but don't get mad at us. we didn't draw any conclusions.)

the river

questions aren't enough. an essay has to come up with answers. they don't always, of course. sometimes you start with a promising question and get nowhere. but those you don't publish. those are like experiments that get inconclusive results. an essay you publish ought to tell the reader something he didn't already know.

but *what* you tell him doesn't matter, so long as it's interesting. i'm sometimes accused of meandering. in defend-a-position writing that would be a flaw. there you're not concerned with truth. you already know where you're going, and you want to go straight there, blustering through obstacles, and hand-waving your way across swampy ground. but that's not what you're trying to do in an essay. an essay is supposed to be a search for truth. it would be suspicious if it didn't meander.

the meander (aka menderes) is a river in turkey. as you might expect, it winds all over the place. but it doesn't do this out of frivolity. the path it has discovered is the most economical route to the sea.[6]

the river's algorithm is simple. at each step, flow down. for the essayist this translates to: flow interesting. of all the places to go next, choose the most interesting. one can't have quite as little foresight as a river. i always know generally what i want to write about. but not the specific conclusions i want to reach; from paragraph to paragraph i let the ideas take their course.

this doesn't always work. sometimes, like a river, one runs up against a wall. then i do the same thing the river does: backtrack. at one point in this

essay i found that after following a certain thread i ran out of ideas. i had to go back seven paragraphs and start over in another direction.
fundamentally an essay is a train of thought-- but a cleaned-up train of thought, as dialogue is cleaned-up conversation. real thought, like real conversation, is full of false starts. it would be exhausting to read. you need to cut and fill to emphasize the central thread, like an illustrator inking over a pencil drawing. but don't change so much that you lose the spontaneity of the original.
err on the side of the river. an essay is not a reference work. it's not something you read looking for a specific answer, and feel cheated if you don't find it. i'd much rather read an essay that went off in an unexpected but interesting direction than one that plodded dutifully along a prescribed course.

surprise

so what's interesting? for me, interesting means surprise. interfaces, as geoffrey james has said, should follow the principle of least astonishment. a button that looks like it will make a machine stop should make it stop, not speed up. essays should do the opposite. essays should aim for maximum surprise.
i was afraid of flying for a long time and could only travel vicariously. when friends came back from faraway places, it wasn't just out of politeness that i asked what they saw. i really wanted to know. and i found the best way to get information out of them was to ask what surprised them. how was the place different from what they expected? this is an extremely useful question. you can ask it of the most unobservant people, and it will extract information they didn't even know they were recording.
surprises are things that you not only didn't know, but that contradict things you thought you knew. and so they're the most valuable sort of fact you can get. they're like a food that's not merely healthy, but counteracts the unhealthy effects of things you've already eaten.
how do you find surprises? well, therein lies half the work of essay writing. (the other half is expressing yourself well.) the trick is to use yourself as a

proxy for the reader. you should only write about things you've thought about a lot. and anything you come across that surprises you, who've thought about the topic a lot, will probably surprise most readers.
for example, in a recent essay i pointed out that because you can only judge computer programmers by working with them, no one knows who the best programmers are overall. i didn't realize this when i began that essay, and even now i find it kind of weird. that's what you're looking for.
so if you want to write essays, you need two ingredients: a few topics you've thought about a lot, and some ability to ferret out the unexpected.
what should you think about? my guess is that it doesn't matter-- that anything can be interesting if you get deeply enough into it. one possible exception might be things that have deliberately had all the variation sucked out of them, like working in fast food. in retrospect, was there anything interesting about working at baskin-robbins? well, it was interesting how important color was to the customers. kids a certain age would point into the case and say that they wanted yellow. did they want french vanilla or lemon? they would just look at you blankly. they wanted yellow. and then there was the mystery of why the perennial favorite pralines 'n' cream was so appealing. (i think now it was the salt.)
and the difference in the way fathers and mothers bought ice cream for their kids: the fathers like benevolent kings bestowing largesse, the mothers harried, giving in to pressure. so, yes, there does seem to be some material even in fast food.
i didn't notice those things at the time, though. at sixteen i was about as observant as a lump of rock. i can see more now in the fragments of memory i preserve of that age than i could see at the time from having it all happening live, right in front of me.

observation

so the ability to ferret out the unexpected must not merely be an inborn one. it must be something you can learn. how do you learn it?
to some extent it's like learning history. when you first read history, it's just a whirl of names and dates. nothing seems to stick. but the more you learn,

the more hooks you have for new facts to stick onto-- which means you accumulate knowledge at an exponential rate. once you remember that normans conquered england in 1066, it will catch your attention when you hear that other normans conquered southern italy at about the same time. which will make you wonder about normandy, and take note when a third book mentions that normans were not, like most of what is now called france, tribes that flowed in as the roman empire collapsed, but vikings (norman = north man) who arrived four centuries later in 911. which makes it easier to remember that dublin was also established by vikings in the 840s. etc, etc squared.

collecting surprises is a similar process. the more anomalies you've seen, the more easily you'll notice new ones. which means, oddly enough, that as you grow older, life should become more and more surprising. when i was a kid, i used to think adults had it all figured out. i had it backwards. kids are the ones who have it all figured out. they're just mistaken.

when it comes to surprises, the rich get richer. but (as with wealth) there may be habits of mind that will help the process along. it's good to have a habit of asking questions, especially questions beginning with why. but not in the random way that three year olds ask why. there are an infinite number of questions. how do you find the fruitful ones?

i find it especially useful to ask why about things that seem wrong. for example, why should there be a connection between humor and misfortune? why do we find it funny when a character, even one we like, slips on a banana peel? there's a whole essay's worth of surprises there for sure.

if you want to notice things that seem wrong, you'll find a degree of skepticism helpful. i take it as an axiom that we're only achieving 1% of what we could. this helps counteract the rule that gets beaten into our heads as children: that things are the way they are because that is how things have to be. for example, everyone i've talked to while writing this essay felt the same about english classes-- that the whole process seemed pointless. but none of us had the balls at the time to hypothesize that it was, in fact, all a mistake. we all thought there was just something we weren't getting.

i have a hunch you want to pay attention not just to things that seem wrong, but things that seem wrong in a humorous way. i'm always pleased when i see someone laugh as they read a draft of an essay. but why should i be? i'm aiming for good ideas. why should good ideas be funny? the connection may be surprise. surprises make us laugh, and surprises are what one wants to deliver.

i write down things that surprise me in notebooks. i never actually get around to reading them and using what i've written, but i do tend to reproduce the same thoughts later. so the main value of notebooks may be what writing things down leaves in your head.

people trying to be cool will find themselves at a disadvantage when collecting surprises. to be surprised is to be mistaken. and the essence of cool, as any fourteen year old could tell you, is *nil admirari.* when you're mistaken, don't dwell on it; just act like nothing's wrong and maybe no one will notice.

one of the keys to coolness is to avoid situations where inexperience may make you look foolish. if you want to find surprises you should do the opposite. study lots of different things, because some of the most interesting surprises are unexpected connections between different fields. for example, jam, bacon, pickles, and cheese, which are among the most pleasing of foods, were all originally intended as methods of preservation. and so were books and paintings.

whatever you study, include history-- but social and economic history, not political history. history seems to me so important that it's misleading to treat it as a mere field of study. another way to describe it is *all the data we have so far.*

among other things, studying history gives one confidence that there are good ideas waiting to be discovered right under our noses. swords evolved during the bronze age out of daggers, which (like their flint predecessors) had a hilt separate from the blade. because swords are longer the hilts kept breaking off. but it took five hundred years before someone thought of casting hilt and blade as one piece.

disobedience

above all, make a habit of paying attention to things you're not supposed to, either because they're "inappropriate," or not important, or not what you're supposed to be working on. if you're curious about something, trust your instincts. follow the threads that attract your attention. if there's something you're really interested in, you'll find they have an uncanny way of leading back to it anyway, just as the conversation of people who are especially proud of something always tends to lead back to it.

for example, i've always been fascinated by comb-overs, especially the extreme sort that make a man look as if he's wearing a beret made of his own hair. surely this is a lowly sort of thing to be interested in-- the sort of superficial quizzing best left to teenage girls. and yet there is something underneath. the key question, i realized, is how does the comber-over not see how odd he looks? and the answer is that he got to look that way *incrementally.* what began as combing his hair a little carefully over a thin patch has gradually, over 20 years, grown into a monstrosity. gradualness is very powerful. and that power can be used for constructive purposes too: just as you can trick yourself into looking like a freak, you can trick yourself into creating something so grand that you would never have dared to *plan* such a thing. indeed, this is just how most good software gets created. you start by writing a stripped-down kernel (how hard can it be?) and gradually it grows into a complete operating system. hence the next leap: could you do the same thing in painting, or in a novel?

see what you can extract from a frivolous question? if there's one piece of advice i would give about writing essays, it would be: don't do as you're told. don't believe what you're supposed to. don't write the essay readers expect; one learns nothing from what one expects. and don't write the way they taught you to in school.

the most important sort of disobedience is to write essays at all. fortunately, this sort of disobedience shows signs of becoming rampant. it used to be that only a tiny number of officially approved writers were allowed to write essays. magazines published few of them, and judged them less by

what they said than who wrote them; a magazine might publish a story by an unknown writer if it was good enough, but if they published an essay on x it had to be by someone who was at least forty and whose job title had x in it. which is a problem, because there are a lot of things insiders can't say precisely because they're insiders.

the internet is changing that. anyone can publish an essay on the web, and it gets judged, as any writing should, by what it says, not who wrote it.

who are you to write about x? you are whatever you wrote.

popular magazines made the period between the spread of literacy and the arrival of tv the golden age of the short story. the web may well make this the golden age of the essay. and that's certainly not something i realized when i started writing this.

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notes

1 i'm thinking of oresme (c. 1323-82). but it's hard to pick a date, because there was a sudden drop-off in scholarship just as europeans finished assimilating classical science. the cause may have been the plague of 1347; the trend in scientific progress matches the population curve.

2 parker, william r. "where do college english departments come from?" *college english* 28 (1966-67), pp. 339-351. reprinted in gray, donald j. (ed). *the department of english at indiana university bloomington 1868-1970.* indiana university publications.

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3 i'm compressing the story a bit. at first literature took a back seat to philology, which (a) seemed more serious and (b) was popular in germany, where many of the leading scholars of that generation had been trained.

in some cases the writing teachers were transformed *in situ* into english professors. francis james child, who had been boylston professor of rhetoric at harvard since 1851, became in 1876 the university's first professor of english.

4 parker, *op. cit.*, p. 25.

5 the undergraduate curriculum or *trivium* (whence "trivial") consisted of latin grammar, rhetoric, and logic. candidates for masters' degrees went on to study the *quadrivium* of arithmetic, geometry, music, and astronomy. together these were the seven liberal arts.

the study of rhetoric was inherited directly from rome, where it was considered the most important subject. it would not be far from the truth to say that education in the classical world meant training landowners' sons to speak well enough to defend their interests in political and legal disputes.

6 trevor blackwell points out that this isn't strictly true, because the outside edges of curves erode faster.

thanks to ken anderson, trevor blackwell, sarah harlin, jessica livingston, jackie mcdonough, and robert morris for reading drafts of this.

11

what you (want to)* want

november 2022
paulgraham.com/want.html

since i was about 9 i've been puzzled by the apparent contradiction between being made of matter that behaves in a predictable way, and the feeling that i could choose to do whatever i wanted. at the time i had a self-interested motive for exploring the question. at that age (like most succeeding ages) i was always in trouble with the authorities, and it seemed to me that there might possibly be some way to get out of trouble by arguing that i wasn't responsible for my actions. i gradually lost hope of that, but the puzzle remained: how do you reconcile being a machine made of matter with the feeling that you're free to choose what you do? [1]

the best way to explain the answer may be to start with a slightly wrong version, and then fix it. the wrong version is: you can do what you want, but you can't want what you want. yes, you can control what you do, but you'll do what you want, and you can't control that.

the reason this is mistaken is that people do sometimes change what they want. people who don't want to want something — drug addicts, for example — can sometimes make themselves stop wanting it. and people who want to want something — who want to like classical music, or broccoli — sometimes succeed.

so we modify our initial statement: you can do what you want, but you can't want to want what you want.

that's still not quite true. it's possible to change what you want to want. i can imagine someone saying "i decided to stop wanting to like classical

music." but we're getting closer to the truth. it's rare for people to change what they want to want, and the more "want to"s we add, the rarer it gets. we can get arbitrarily close to a true statement by adding more "want to"s in much the same way we can get arbitrarily close to 1 by adding more 9s to a string of 9s following a decimal point. in practice three or four "want to"s must surely be enough. it's hard even to envision what it would mean to change what you want to want to want to want, let alone actually do it. so one way to express the correct answer is to use a regular expression. you can do what you want, but there's some statement of the form "you can't (want to)* want what you want" that's true. ultimately you get back to a want that you don't control. [2]

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notes

1 i didn't know when i was 9 that matter might behave randomly, but i don't think it affects the problem much. randomness destroys the ghost in the machine as effectively as determinism.

2 if you don't like using an expression, you can make the same point using higher-order desires: there is some n such that you don't control your nth-order desires.

thanks to trevor blackwell, jessica livingston, robert morris, and michael nielsen for reading drafts of this.

12

how to get new ideas

january 2023
paulgraham.com/getideas.html

(someone fed my essays into gpt to make something that could answer questions based on them, then asked it where good ideas come from. the answer was ok, but not what i would have said. this is what i would have said.)
the way to get new ideas is to notice anomalies: what seems strange, or missing, or broken? you can see anomalies in everyday life (much of standup comedy is based on this), but the best place to look for them is at the frontiers of knowledge.
knowledge grows fractally. from a distance its edges look smooth, but when you learn enough to get close to one, you'll notice it's full of gaps. these gaps will seem obvious; it will seem inexplicable that no one has tried x or wondered about y. in the best case, exploring such gaps yields whole new fractal buds.